

MONOTONICITY OF A RATIO OF COMPLETE HOMOGENEOUS SYMMETRIC POLYNOMIALS

VAMSHI JANDHYALA

ABSTRACT. For the complete homogeneous symmetric polynomial h_ℓ in n variables, write $H_i(x) = h_\ell(1^{[n-i]}, x^{[i]})$ for its value when i of the variables equal x and the rest equal 1. Ait-Haddou conjectured that $\Psi_{i,n} = H_i^2/(H_{i+1}H_{i-1})$ is strictly increasing on $(1, \infty)$ for $2 \leq i \leq n-2$. We prove this for every $\ell \geq 1$ and every $1 \leq i \leq n-1$. Treating the number i of x 's as a continuous variable t , the logarithmic derivative of $\Psi_{i,n}$ becomes a positive multiple of a second difference in t of a ratio of two consecutive polynomials of a Meixner family, and that ratio is convex because their zeros interlace.

Status of this preprint. This version has not yet been independently reviewed.

1. INTRODUCTION

Take a symmetric polynomial in n variables, set every variable to 1, and then move i of them to a common value x . How does the value depend on i ? For the complete homogeneous symmetric polynomials Ait-Haddou asked whether a measure of curvature in i , the ratio $\Psi_{i,n}$ below, strictly increases as x moves above 1. This note shows that it does; since $\Psi_{i,n}(1) = 1$, it follows that the values are strictly log-concave in i for every positive $x \neq 1$.

The complete homogeneous symmetric polynomial of degree ℓ in n variables is

$$h_\ell(x_1, \dots, x_n) = \sum_{1 \leq j_1 \leq j_2 \leq \dots \leq j_\ell \leq n} x_{j_1} x_{j_2} \cdots x_{j_\ell}.$$

Write $x^{[k]}$ for x repeated k times, and for $0 \leq i \leq n$ put

$$H_i(x) = h_\ell(1^{[n-i]}, x^{[i]}).$$

In 2024 Ait-Haddou asked on MathOverflow [9] whether, for every $\ell \geq 1$, each of the functions

$$\Psi_{i,n}(x) = \frac{H_i(x)^2}{H_{i+1}(x)H_{i-1}(x)}, \quad 2 \leq i \leq n-2,$$

is strictly increasing on $(1, \infty)$, and reported a proof for $\Psi_{2,4}$. The question received two bounties and no answer until [10]. It belongs to a line of work on monotonicity and inequalities for ratios of symmetric polynomials [3, 11, 1, 6, 13, 8]; here a single polynomial h_ℓ is compared at three neighbouring points.

A first case. For $\ell = 1$, $H_i = n + iy$ with $y = x - 1$, so

$$\Psi_{i,n} = \frac{(n + iy)^2}{(n + iy)^2 - y^2} = \left(1 - \frac{y^2}{(n + iy)^2}\right)^{-1},$$

and $y/(n + iy)$ increases with $y > 0$. For larger ℓ each H_i has degree ℓ in x and no such simplification is available; the proof below works instead with i as the variable.

Theorem 1. *For all integers $n \geq 2$, $\ell \geq 1$ and $1 \leq i \leq n-1$, the function $\Psi_{i,n}$ is strictly increasing on $(1, \infty)$.*

Date: Preprint, version 1, 9 October 2026. Not peer reviewed.

2020 Mathematics Subject Classification. 05E05, 26D05, 33C45.

Key words and phrases. complete homogeneous symmetric polynomial, Turán-type inequality, Meixner polynomials, interlacing.

Since every $H_i(1) = \binom{n+\ell-1}{\ell}$, $\Psi_{i,n}(1) = 1$, so $H_i^2 > H_{i+1}H_{i-1}$ for $x > 1$. Homogeneity gives $H_i(x) = x^\ell h_\ell(1^{[i]}, (1/x)^{[n-i]})$, hence $\Psi_{i,n}(x) = \Psi_{n-i,n}(1/x)$, and each $\Psi_{i,n}$ is strictly decreasing on $(0, 1)$. So $x = 1$ is the minimum and the interval $(1, \infty)$ in the question is the natural one.

Outline. Section 2 interpolates $i \mapsto H_i$ by a polynomial $P_\ell(t)$ of degree ℓ in a real variable t . Section 3 differentiates in x : one binomial identity turns $(x-1)\frac{d}{dx}\log \Psi_{i,n}$ into a positive multiple of the second difference $f(i-1) + f(i+1) - 2f(i)$ of $f = P_{\ell-1}/P_\ell$. Section 4 shows that the P_ℓ satisfy a three-term recurrence of orthogonal type, so their zeros are real, negative and interlacing, and f is a sum of positive multiples of $1/(t-r)$ with poles $r < 0$. Such a function is convex on $[0, \infty)$ (Figure 1), which proves the theorem.

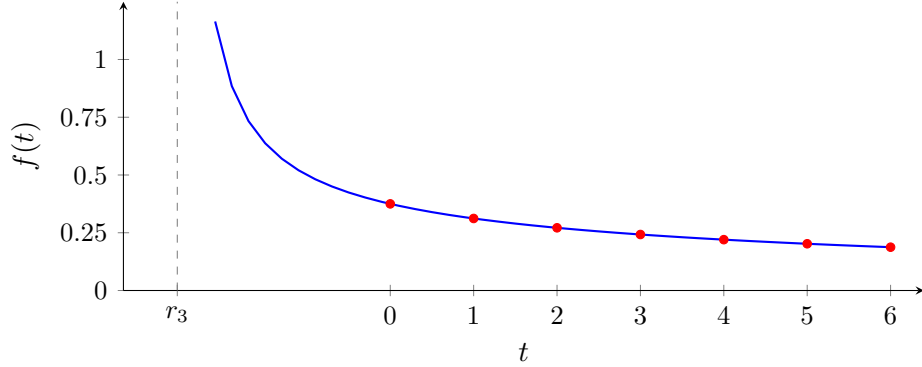


FIGURE 1. The ratio $f = P_2/P_3$ (the case $\ell = 3$) for $n = 6$ and $x = 2$. The zeros of P_3 are $r_1 \approx -16.45$, $r_2 = -8$ and $r_3 \approx -2.55$ (dashed). To the right of r_3 the curve is convex, so the second differences of f at the integer points $t = 0, \dots, 6$ (red) are positive, and Proposition 3 turns that into $\Psi'_{i,6} > 0$ for $1 \leq i \leq 5$.

Throughout, $x > 1$ is fixed and $y = x - 1 > 0$. All statements about P_ℓ hold for real $n > 0$, although only integer n is needed for the theorem.

2. THE NUMBER OF x 'S AS A VARIABLE

The function $\Psi_{i,n}$ compares H at the three integers $i-1, i, i+1$. To compare them through a derivative and a convexity argument we need H_i for real i , which the following formula supplies.

For real t and integer $k \geq 0$ write $\binom{t+k-1}{k} = t(t+1)\cdots(t+k-1)/k!$, a polynomial of degree k in t . Define

$$(1) \quad P_\ell(t) = \sum_{k=0}^{\ell} \binom{n+\ell-1}{\ell-k} \binom{t+k-1}{k} y^k \quad (\ell \geq 0).$$

In words, $P_\ell(t)$ is a sum of rising factorials in t with coefficients that are polynomials in y ; the number t of x 's enters only through those rising factorials.

Lemma 1. P_ℓ is a polynomial in t of degree ℓ with leading coefficient $y^\ell/\ell!$, positive on $[0, \infty)$, and $P_\ell(i) = H_i(x)$ for $i = 0, 1, \dots, n$. As formal power series in z , for every real t ,

$$(2) \quad \sum_{\ell \geq 0} P_\ell(t) z^\ell = (1-z)^{-(n-t)} (1-xz)^{-t}.$$

Proof. The degree and leading coefficient come from the term $k = \ell$. For $t \geq 0$ every term of (1) is non-negative and the term $k = 0$ equals $\binom{n+\ell-1}{\ell} > 0$. For (2), write $1 - xz = (1-z)(1-yw)$ with $w = z/(1-z)$, so that the right side equals $(1-z)^{-n}(1-yw)^{-t} = \sum_k \binom{t+k-1}{k} y^k z^k (1-z)^{-n-k}$, and the coefficient of z^ℓ in $z^k(1-z)^{-n-k}$ is $\binom{n+\ell-1}{\ell-k}$. At $t = i$

the right side of (2) is $\prod_j (1 - x_j z)^{-1}$ for $(x_j) = (1^{[n-i]}, x^{[i]})$, the generating function of h_ℓ , so $P_\ell(i) = H_i(x)$. $\square$

3. THE REDUCTION TO A SECOND DIFFERENCE

Lemma 2. For $\ell \geq 1$, $y \frac{\partial}{\partial y} P_\ell(t) = \ell P_\ell(t) - (n + \ell - 1) P_{\ell-1}(t)$.

Proof. The operator $y \partial / \partial y$ multiplies the k -th term of (1) by k . With $N = n + \ell - 1$,

$$k \binom{N}{\ell - k} = \ell \binom{N}{\ell - k} - (\ell - k) \binom{N}{\ell - k} = \ell \binom{N}{\ell - k} - N \binom{N-1}{\ell - 1 - k},$$

and $\binom{N-1}{(\ell-1)-k} = \binom{n+(\ell-1)-1}{(\ell-1)-k}$ is the coefficient in $P_{\ell-1}$ (the term $k = \ell$ has no counterpart there, and $\binom{N-1}{-1} = 0$). $\square$

Proposition 3. Let $f = P_{\ell-1}/P_\ell$ on $[0, \infty)$. For $\ell \geq 1$, $1 \leq i \leq n-1$ and $x > 1$,

$$(x-1) \frac{d}{dx} \log \Psi_{i,n}(x) = (n + \ell - 1) [f(i-1) + f(i+1) - 2f(i)].$$

Proof. Since $d/dx = d/dy$, dividing Lemma 2 by $P_\ell(t) > 0$ gives $(x-1) \frac{d}{dx} \log H_t = \ell - (n + \ell - 1)f(t)$ at $t = i-1, i, i+1$. In $\log \Psi_{i,n} = 2 \log H_i - \log H_{i+1} - \log H_{i-1}$ the constant ℓ cancels. $\square$

So Theorem 1 follows once f is strictly convex on $[0, \infty)$.

4. RECURRENCE, INTERLACING AND CONVEXITY

Lemma 4. $P_0 = 1$, $P_1(t) = n + yt$, and for $\ell \geq 1$

$$(3) \quad (\ell + 1)P_{\ell+1} = ((1+x)\ell + n + yt)P_\ell - x(n + \ell - 1)P_{\ell-1}.$$

Proof. Let $F(z)$ be the right side of (2). Logarithmic differentiation gives $zF'/F = z(\frac{n-t}{1-z} + \frac{tx}{1-xz})$, that is,

$$(1-z)(1-xz)zF'(z) = z((n+yt) - nxz)F(z).$$

The coefficient of $z^{\ell+1}$ on the left is $(\ell+1)P_{\ell+1} - (1+x)\ell P_\ell + x(\ell-1)P_{\ell-1}$ and on the right it is $(n+yt)P_\ell - nxP_{\ell-1}$. $\square$

In (3), $P_{\ell+1}$ is P_ℓ times a linear factor in t with positive slope y , minus a positive multiple of $P_{\ell-1}$: the shape of the recurrence of orthogonal polynomials. Indeed, comparing (1) with the hypergeometric form of the Meixner polynomials [7, (9.10.1)] gives $P_\ell(t) = \binom{n+\ell-1}{\ell} M_\ell(-t; n, 1/x)$. The next lemma is the standard consequence [12, §§3.3–3.4], [2], proved here in the form needed.

Lemma 5. Let $D_\ell = P'_\ell P_{\ell-1} - P_\ell P'_{\ell-1}$, derivatives in t . Then $D_1 = y$ and

$$(4) \quad (\ell + 1)D_{\ell+1} = y P_\ell^2 + x(n + \ell - 1)D_\ell,$$

so $D_\ell > 0$ on $\mathbb{R}$ for every $\ell \geq 1$. Consequently P_ℓ has ℓ simple zeros, all negative, and at each zero r of P_ℓ the number $B_r = P_{\ell-1}(r)/P'_\ell(r)$ is positive.

Proof. Differentiate (3) in t , multiply by P_ℓ , and subtract (3) multiplied by P'_ℓ ; the terms in $P_\ell P'_\ell$ cancel and (4) remains. Induction gives $D_\ell > 0$.

The zeros are real, simple and negative by induction on ℓ ; P_1 has the single zero $-n/y < 0$. Suppose P_ℓ has zeros $r_1 < \dots < r_\ell < 0$. At r_j , (4) reads $(\ell+1)D_{\ell+1}(r_j) = -(\ell+1)P_{\ell+1}(r_j)P'_\ell(r_j)$, which is positive. The simple zeros of P_ℓ alternate the sign of P'_ℓ , which is $(-1)^{\ell-j}$ at r_j because the leading coefficient is positive. So $P_{\ell+1}(r_j)$ has sign $(-1)^{\ell-j+1}$. Together with $P_{\ell+1}(0) > 0$ (Lemma 1) and the sign $(-1)^{\ell+1}$ of $P_{\ell+1}$ near $-\infty$, these are $\ell+2$ points with alternating signs, so $P_{\ell+1}$ has a zero in each of the $\ell+1$ gaps, all in $(-\infty, 0)$. Finally, at a zero r of P_ℓ , $D_\ell(r) = P'_\ell(r)P_{\ell-1}(r) > 0$, so $B_r > 0$. $\square$

Proof of Theorem 1. By Lemma 5, $\deg P_{\ell-1} < \deg P_\ell$ and the zeros of P_ℓ are simple, so partial fractions give

$$f(t) = \frac{P_{\ell-1}(t)}{P_\ell(t)} = \sum_r \frac{B_r}{t-r}, \quad B_r > 0, \quad r < 0.$$

For $t \geq 1$ each $a = t - r$ exceeds 1, and

$$\frac{1}{a-1} + \frac{1}{a+1} - \frac{2}{a} = \frac{2}{a(a^2-1)} > 0.$$

This is the convexity visible in Figure 1. Summing with the weights B_r gives $f(t-1) + f(t+1) - 2f(t) > 0$ for all $t \geq 1$, in particular at $t = i$ with $1 \leq i \leq n-1$. By Proposition 3, $\Psi'_{i,n}(x) > 0$ for every $x > 1$. $\square$

5. REMARKS

Remark 1. The proof shows more than was asked: $t \mapsto P_{\ell-1}(t)/P_\ell(t)$ is convex on $(r_{\max}, \infty) \supset [0, \infty)$ for every real $n > 0$, so the conclusion holds with n real and i any real number in $[1, n-1]$, with H_t defined by (1). The range $1 \leq i \leq n-1$ is the largest for which $\Psi_{i,n}$ is defined.

Remark 2. The inequality $\Psi_{i,n} > 1$ on $(1, \infty)$ says that $P_\ell(t)^2 - P_\ell(t-1)P_\ell(t+1) > 0$ at integer t , a Turán-type inequality for one Meixner polynomial at shifted values of its variable. Karlin and Szegő [5] studied the sign of Hankel determinants of discrete orthogonal polynomials, including Meixner, across consecutive degrees, and Durán [4] found identities for their Casorati determinants; we have not found the inequality above, or the monotonicity in x , in that literature.

Remark 3. The interlacing in Lemma 5 and the positivity of the weights B_r are classical for orthogonal polynomials; the new step is Proposition 3, which turns the derivative in the parameter x into a difference in the variable t .

Verification. The identities of Lemmas 1–5 and Proposition 3 were checked in exact arithmetic for small n and ℓ , and the theorem on grids of rational points, by a script that also confirms the example $\ell = 1$, the symmetry $\Psi_{i,n}(x) = \Psi_{n-i,n}(1/x)$ and the data of Figure 1, and rejects deliberately altered statements. It is at

<https://github.com/jvbk/mathematics/tree/main/symmetric-polynomial-ratio>.

ACKNOWLEDGEMENTS

The proof, text and verification program were developed with the AI assistant Claude (Anthropic), under the author's direction. The author thanks Rachid Ait-Haddou for the question. The author is responsible for the contents.

REFERENCES

- [1] R. Ait-Haddou and M.-L. Mazure, *The fundamental blossoming inequality in Chebyshev spaces I: applications to Schur functions*, Found. Comput. Math. **18** (2018), 135–158, doi:10.1007/s10208-016-9334-8.
- [2] T. S. Chihara, *An Introduction to Orthogonal Polynomials*, Gordon and Breach, New York, 1978.
- [3] A. Cuttler, C. Greene and M. Skandera, *Inequalities for symmetric means*, European J. Combin. **32** (2011), 745–761, doi:10.1016/j.ejc.2011.01.020.
- [4] A. J. Durán, *Symmetries for Casorati determinants of classical discrete orthogonal polynomials*, Proc. Amer. Math. Soc. **142** (2014), 915–930, doi:10.1090/S0002-9939-2013-11802-2.
- [5] S. Karlin and G. Szegő, *On certain determinants whose elements are orthogonal polynomials*, J. Analyse Math. **8** (1960/61), 1–157.
- [6] A. Khare and T. Tao, *On the sign patterns of entrywise positivity preservers in fixed dimension*, Amer. J. Math. **143** (2021), 1863–1929, doi:10.1353/ajm.2021.0049.
- [7] R. Koekoek, P. A. Lesky and R. F. Swarttouw, *Hypergeometric Orthogonal Polynomials and Their q -Analogues*, Springer, Berlin, 2010.
- [8] C. McSwiggen and S. Sahi, *Majorization inequalities from logarithmic convexity*, preprint (2026), arXiv:2605.12680.

- [9] R. Ait-Haddou, *Monotonicity of ratio of symmetric polynomials*, MathOverflow question 467261 (18 March 2024), <https://mathoverflow.net/q/467261>, accessed 9 October 2026.
- [10] V. Jandhyala, answer to MathOverflow question 467261 (7 October 2026), <https://mathoverflow.net/a/515799>.
- [11] S. Sra, *On inequalities for normalized Schur functions*, European J. Combin. **51** (2016), 492–494, doi:10.1016/j.ejc.2015.07.005.
- [12] G. Szegő, *Orthogonal Polynomials*, 4th ed., Amer. Math. Soc. Colloq. Publ. 23, Providence, RI, 1975.
- [13] J. Xu and Y. Yao, *Majorization and inequalities among complete homogeneous symmetric functions*, European J. Combin. **136** (2026), 104389, doi:10.1016/j.ejc.2026.104389.